

Math question bank — Grade 10–11

63 questions · 7 topics × (3 easy + 3 medium + 3 challenge) · For review before database seeding · Source: reviewed_generated

1. Quadratic Equations

Easy

#	Question	Answer
1	Solve: $x^2 = 49$	$x = 7$ or $x = -7$
2	Solve: $x^2 - 5x = 0$	$x = 0$ or $x = 5$
3	Solve: $x^2 - 9 = 0$	$x = 3$ or $x = -3$

Medium

#	Question	Answer
4	Solve: $x^2 - 7x + 12 = 0$	$x = 3$ or $x = 4$
5	Solve: $x^2 + 2x - 15 = 0$	$x = -5$ or $x = 3$
6	Find the discriminant of $x^2 - 6x + 9 = 0$. What does it tell you about the roots?	0; roots are real and equal

Challenge

#	Question	Answer
7	Solve: $2x^2 - 5x - 3 = 0$	$x = 3$ or $x = -1/2$
8	Without solving, find the sum and product of the roots of $x^2 - 8x + 5 = 0$.	Sum = 8, product = 5
9	The product of two consecutive positive integers is 156. Find the integers.	12 and 13

2. Systems of Equations

Easy

#	Question	Answer
10	Solve: $x + y = 10$ and $x - y = 4$	$x = 7, y = 3$
11	Solve: $y = 2x$ and $x + y = 12$	$x = 4, y = 8$
12	Solve: $2x + y = 7$ and $y = 1$	$x = 3, y = 1$

Medium

#	Question	Answer
13	Solve: $2x + 3y = 12$ and $x - y = 1$	$x = 3, y = 2$
14	Solve: $x + 2y = 8$ and $2x - y = 6$	$x = 4, y = 2$

15	Two pens and three pencils cost 24. One pen and two pencils cost 14. Find the cost of one pen and one pencil.	Pen = 6, pencil = 4
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Challenge

#	Question	Answer
16	Solve: $3x - 2y = 5$ and $2x + 5y = 16$	$x = 3, y = 2$
17	The digits of a two-digit number add up to 11. Reversing the digits gives a number 27 more than the original. Find the number.	47
18	A boat travels 30 km downstream in 2 hours and returns upstream in 3 hours. Find the speed of the boat in still water and the speed of the stream.	Boat = 12.5 km/h, stream = 2.5 km/h

3. Trigonometry

Easy

#	Question	Answer
19	Find the value of $\sin 30^\circ$.	$1/2$
20	Find the value of $\tan 45^\circ$.	1
21	Find the value of $\cos 60^\circ$.	$1/2$

Medium

#	Question	Answer
22	If $\sin \theta = 3/5$ and θ is acute, find $\cos \theta$.	$4/5$
23	Evaluate: $\sin^2 30^\circ + \cos^2 30^\circ$	1
24	Evaluate: $\tan 30^\circ \times \tan 60^\circ$	1

Challenge

#	Question	Answer
25	From a point 60 m from the base of a tower, the angle of elevation of the top is 30° . Find the height of the tower.	$20\sqrt{3}$ m (≈ 34.6 m)
26	If $\tan \theta = 1$ and θ is acute, find $\sin \theta + \cos \theta$.	$\sqrt{2}$
27	Evaluate: $\sin 60^\circ \cos 30^\circ + \cos 60^\circ \sin 30^\circ$	1

4. Coordinate Geometry

Easy

#	Question	Answer
28	Find the distance between (0, 0) and (3, 4).	5
29	In which quadrant does the point $(-2, 5)$ lie?	Second quadrant
30	Find the midpoint of (2, 4) and (6, 8).	$(4, 6)$

Medium

#	Question	Answer
31	Find the distance between (1, 2) and (7, 10).	10
32	Find the slope of the line through (2, 3) and (6, 11).	2
33	Find the point on the x-axis equidistant from (2, -5) and (-2, 9).	$(-7, 0)$

Challenge

#	Question	Answer
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34	A triangle has vertices $(0, 0)$, $(4, 0)$ and $(4, 3)$. Find its perimeter.	12
35	The distance between $(3, y)$ and $(3, -2)$ is 7. Find the possible values of y .	$y = 5$ or $y = -9$
36	In what ratio does the point $(2, 5)$ divide the line segment from $(1, 3)$ to $(4, 9)$?	1 : 2

5. Functions & Graphs

Easy

#	Question	Answer
37	If $f(x) = 2x + 3$, find $f(4)$.	11
38	If $f(x) = x^2$, find $f(-3)$.	9
39	What is the y-intercept of the line $y = 3x - 2$?	-2

Medium

#	Question	Answer
40	If $f(x) = x^2 - 2x$, find $f(5)$.	15
41	State the slope and y-intercept of $y = -2x + 7$.	Slope = -2, y-intercept = 7
42	If $f(x) = 3x - 1$ and $f(a) = 14$, find a .	$a = 5$

Challenge

#	Question	Answer
43	If $f(x) = 2x + 1$ and $g(x) = x^2$, find $f(g(3))$.	19
44	State the domain of $f(x) = 1/(x - 4)$.	All real numbers except $x = 4$
45	If $f(x) = 2^x$, find $f(0) + f(1) + f(2) + f(3)$.	15

6. Sequences & Series

Easy

#	Question	Answer
46	Find the next term: 3, 7, 11, 15, ...	19
47	Find the next term: 2, 6, 18, 54, ...	162
48	Find the common difference of 12, 9, 6, 3, ...	-3

Medium

#	Question	Answer
49	Find the 15th term of the arithmetic sequence 7, 10, 13, ...	49
50	Find the sum of the first 20 natural numbers.	210
51	Which term of the sequence 5, 8, 11, ... is 62?	20th term

Challenge

#	Question	Answer
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52	Find the sum of the first 15 terms of an arithmetic sequence with first term 3 and common difference 5.	570
53	Three numbers in arithmetic progression add up to 27, and the product of the first and last is 65. Find the numbers.	5, 9, 13
54	In a geometric sequence the first term is 2 and the 4th term is 54. Find the common ratio.	3

7. Statistics

Easy

#	Question	Answer
55	Find the mean of 6, 8, 10, 12, 14.	10
56	Find the mode of 2, 3, 3, 4, 5, 3.	3
57	Find the median of 5, 1, 9.	5

Medium

#	Question	Answer
58	Find the mean of the first 10 natural numbers.	5.5
59	The mean of 5 observations is 18. Four of them are 15, 20, 22 and 12. Find the fifth.	21
60	Find the median of 3, 8, 12, 5, 10, 15.	9

Challenge

#	Question	Answer
61	The mean of 20 numbers is 45. It is later found that one number was read as 25 instead of 52. Find the correct mean.	46.35
62	The mean of x , $x + 2$, $x + 4$, $x + 6$ and $x + 8$ is 11. Find x .	$x = 7$
63	Using the relation $\text{mode} = 3(\text{median}) - 2(\text{mean})$, find the mode when the median is 20 and the mean is 18.	24